13. Construct a C program for implementation the various memory allocation strategies.

Code:

main.c

Share Run

```c
#include <stdio.h>
int main() {
    int block[5], process[5];
    int nb, np;
    printf("Enter number of memory blocks: ");
    scanf("%d", &nb);
    printf("Enter block sizes:\n");
    for(int i=0;i<nb;i++)
        scanf("%d",&block[i]);
    printf("Enter number of processes: ");
    scanf("%d",&np);

    printf("Enter process sizes:\n");
    for(int i=0;i<np;i++)
        scanf("%d",&process[i]);

    printf("\nFirst Fit Allocation\n");

    int allocated[10]={0};

    for(int i=0;i<np;i++) {
        for(int j=0;j<nb;j++) {
            if(block[j]>=process[i] && !allocated[j]) {
                printf("Process %d allocated to Block %d\n",i+1,j+1);
                allocated[j]=1;
                break;
            }
        }
    }

    return 0;
}
```

Output:

Output

```
Enter number of memory blocks: 3
Enter block sizes:
100 200 300

Enter number of processes: 2
Enter process sizes:
700 800


First Fit Allocation


=== Code Execution Successful ===
```